Mathematics practice booklet: Grade 3 expectations

Grade 3 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 3 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 3.OA.A.1, 3.OA.A.3, 3.OA.A.4, 3.OA.B.5, 3.OA.D.8, 3.OA.D.9

A worked example

There are 6 trays with 4 date cookies on each tray. How many cookies are there? Write the multiplication fact and a related division fact.

1. 6 groups of 4 is 6 x 4.
2. 6 x 4 = 24, so there are 24 cookies.
3. A related division fact: 24 / 6 = 4.

24 cookies; 6 x 4 = 24 and 24 / 6 = 4

Questions

1. What does 5 x 7 mean? Say it as groups.

2. A box holds 8 pencils. How many pencils are in 4 boxes?

3. 30 stickers are shared equally among 5 friends. How many does each friend get?

4. Find the missing number: 4 x __ = 36

5. Find the missing number: 42 / __ = 6

6. Use a fact you know to find 6 x 8. Hint: 6 x 8 is the same as 8 x 6, and 5 x 8 = 40.

7. Are 3 x 9 and 9 x 3 equal? Which property shows it?

8. Maha has 3 packs of 6 stickers and gives 5 away. How many does she have left?

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9. What is the rule? 4, 8, 12, 16, __ What comes next?

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10. Write a number sentence for: a number times 7 is 56. Use n for the unknown, then solve.

Section 2: Shapes, measuring and space

Standards practised: 3.MD.C.5, 3.MD.C.6, 3.MD.C.7, 3.MD.D.8, 3.G.A.1, 3.G.A.2, 3.MD.A.1, 3.MD.A.2

A worked example

A rectangular majlis rug is 4 metres long and 3 metres wide. What is its area? What is its perimeter?

1. Area is the number of square metres that cover it: 4 x 3 = 12.
2. Perimeter is the distance round the edge: 4 + 3 + 4 + 3 = 14.

Area 12 square metres; perimeter 14 metres

Questions

1. A rectangle is 5 cm long and 2 cm wide. What is its area?

2. A square has sides of 6 cm. What is its perimeter?

3. A rectangle has area 24 square units and a width of 4 units. What is its length?

4. Name a shape with 4 sides and 4 equal sides. Is it always a square?

5. A shape is split into 4 equal parts. What fraction of the whole is each part?

6. The film starts at 4:35 and lasts 40 minutes. When does it end?

7. A rectangle has sides 7 cm and 3 cm. Find its perimeter.

8. Draw two different rectangles that both have an area of 12 square units. What are their sides?

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9. A jug holds 2 litres of juice. Hamad pours out 500 millilitres. How many millilitres are left? (1 litre = 1000 ml)

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10. Which has the larger area: a 5 by 3 rectangle or a 4 by 4 square?

Section 3: Data, graphs and chance

Standards practised: 3.MD.B.3, 3.MD.B.4

A worked example

A bar graph shows books read in a month. Each square stands for 2 books. Salem has 5 squares and Mona has 3 squares. How many more books did Salem read?

1. Salem read 5 x 2 = 10 books.
2. Mona read 3 x 2 = 6 books.
3. 10 - 6 = 4.

4 more books

Questions

1. In a picture graph each star is 5 votes. A drink has 4 stars. How many votes?

2. In the same graph, juice has 6 stars and milk has 4 stars. How many more votes does juice have?

3. A bar graph scale goes up in 10s. A bar ends halfway between 20 and 30. What value does it show?

4. Pencil lengths in cm are 12, 14, 12, 15, 14, 12. How many pencils are 12 cm?

5. Use the lengths in the last question. What is the longest length, and what is the shortest?

6. On a line plot of lengths measured to the nearest half cm, there are 2 dots at 3 and 4 dots at 3 and a half. How many measurements are there?

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7. A survey: 8 children like football, 5 like swimming, 7 like running. How many children in all?

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8. Which sport in that survey is the most popular, and by how many votes is it ahead of the least popular?

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9. A graph uses 1 square for 3 students. 12 students walk to school. How many squares?

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10. Make a tally of the vowels a, e, i, o, u in your full name. Which letter shows the most?

Section 4: Numbers and calculating

Standards practised: 3.NBT.A.1, 3.NBT.A.2, 3.NBT.A.3, 3.NF.A.1, 3.NF.A.2, 3.NF.A.3

A worked example

Round 487 to the nearest ten and to the nearest hundred.

1. Look at the ones digit for the nearest ten: 7 is 5 or more, so round up: 490.
2. Look at the tens digit for the nearest hundred: 8 is 5 or more, so round up: 500.

490 and 500

Questions

1. Round 364 to the nearest ten.

2. Round 751 to the nearest hundred.

3. Add: 346 + 278

4. Subtract: 503 - 217

5. Multiply: 6 x 40

6. A bag holds 125 dates and a second bag holds 238. How many in total?

7. What fraction of the circle is shaded if 3 of 8 equal parts are shaded?

8. Which is greater: 1/2 or 1/4? Explain.

9. Write a fraction equal to 1/2 with 4 as the denominator.

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10. Place 3/4 on a number line from 0 to 1. At which quarter mark do you stop?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 5 groups of 7 (7 + 7 + 7 + 7 + 7)
2. 32
3. 6
4. 9
5. 7
6. 48 (40 + 8)
7. Yes, both are 27: the order does not change the product (commutative property).
8. 13 (3 x 6 = 18, then 18 - 5 = 13)
9. Add 4; the next number is 20.
10. n x 7 = 56, n = 8

Section 2: Shapes, measuring and space

1. 10 square centimetres
2. 24 cm
3. 6 units
4. A rhombus or a square. A rhombus is not always a square (its angles need not be right angles).
5. 1/4
6. 5:15
7. 20 cm
8. For example 3 by 4 and 2 by 6 (also 1 by 12).
9. 1500 ml
10. The 4 by 4 square (16 against 15)

Section 3: Data, graphs and chance

1. 20
2. 10 (30 - 20)
3. 25
4. 3
5. Longest 15 cm, shortest 12 cm
6. 6
7. 20
8. Football, ahead of swimming by 3
9. 4
10. Answers vary with the name.

Section 4: Numbers and calculating

1. 360
2. 800
3. 624
4. 286

5. 240
6. 363
7. 3/8
8. 1/2, because halves are bigger pieces than quarters.
9. 2/4
10. At the third mark of four equal parts, just before 1.